Sequences & Patterns

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Arithmetic sequences

Find the 10th term of the sequence $5, 8, 11, 14, \ldots$

$a = 5$, $d = 3$. $a_{10} = 5 + 9 \times 3 = 32$.

2 [Foundational] • Target: Arithmetic sequences

An arithmetic sequence has first term 4 and common difference 6. Find the 20th term.

$a_{20} = 4 + 19 \times 6 = 4 + 114 = 118$.

3 [Foundational] • Target: Arithmetic sequences

The 5th term of an arithmetic sequence is 23, and the first term is 3. Find the common difference.

$3 + 4d = 23 \Rightarrow 4d = 20 \Rightarrow d = 5$.

4 [Foundational] • Target: Arithmetic sequences

An arithmetic sequence has first term 7 and common difference 5. Which term equals 82?

$7 + (n - 1) \times 5 = 82 \Rightarrow n - 1 = 15 \Rightarrow n = 16$. The 16th term.

5 [Foundational] • Target: Arithmetic sequences

The 8th term of an arithmetic sequence is 34, and the common difference is 4. Find the first term.

$a + 7 \times 4 = 34 \Rightarrow a + 28 = 34 \Rightarrow a = 6$.

6 [Foundational] • Target: Geometric sequences

Find the next term of the sequence $2, 4, 8, 16, \ldots$

Geometric, ratio 2: $16 \times 2 = 32$.

7 [Foundational] • Target: Geometric sequences

A geometric sequence has first term 5 and common ratio 3. Find the 5th term.

$a_5 = 5 \times 3^4 = 5 \times 81 = 405$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

© ClipSAT — clipsat.org | All Rights Reserved

8 [Intermediate] • Target: Geometric sequences

The 4th term of a geometric sequence is 54, and the first term is 2. Find the common ratio.

$2 \times r^3 = 54 \Rightarrow r^3 = 27 \Rightarrow r = 3$.

9 [Intermediate] • Target: Geometric sequences

A geometric sequence has 2nd term 6 and 3rd term 18. Find the first term and the common ratio.

$r = 18 \div 6 = 3$. First term $= 6 \div 3 = 2$.

10 [Intermediate] • Target: Identifying the type of sequence

Is $5, 10, 20, 40, \ldots$ arithmetic, geometric, or neither?

Each term is double the previous one ($r = 2$): geometric.

11 [Intermediate] • Target: Finding missing terms

Find the missing term: $3,\ __,\ 11,\ 15$ (arithmetic).

Common difference $= (15 - 3) \div 3 = 4$. Missing term $= 3 + 4 = 7$.

12 [Intermediate] • Target: Finding missing terms

Find the missing term: $4,\ __,\ 36$ (geometric, all terms positive).

$r = \sqrt{36 \div 4} = \sqrt{9} = 3$. Missing term $= 4 \times 3 = 12$.

13 [Intermediate] • Target: Finding missing terms

Insert two arithmetic means between 5 and 20, forming the four-term sequence $5, __, __, 20$.

Common difference $= (20 - 5) \div 3 = 5$. The sequence is $5, 10, 15, 20$: the missing terms are 10 and 15.

14 [Intermediate] • Target: Finding missing terms

The sequence $x,\ 2x + 1,\ 4x - 1$ is arithmetic. Find x.

Equal consecutive differences: $(2x + 1) - x = (4x - 1) - (2x + 1) \Rightarrow x + 1 = 2x - 2 \Rightarrow x = 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Finding missing terms • Cross-Lesson: 5.01 Quantitative Comparisons

Find the two missing terms of the geometric sequence $2,\ __,\ __,\ 54$.

$r^3 = 54 \div 2 = 27 \Rightarrow r = 3$. The sequence is $2, 6, 18, 54$: the missing terms are 6 and 18.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Sum of an arithmetic series • Cross-Lesson: 7.01 Data Analysis & Probability

Find the sum of the first 10 terms of the sequence $2, 5, 8, 11, \ldots$

$a = 2$, $d = 3$, $n = 10$. $S = \frac{n}{2}(2a + (n-1)d) = 5(4 + 27) = 5 \times 31 = 155$.

17 [Challenging] • Target: Sum of an arithmetic series • Cross-Lesson: 8.01 Word Problems & Reasoning

Find the sum of the first 20 positive integers.

$S = \frac{20 \times 21}{2} = 210$.

18 [Challenging] • Target: Sum of an arithmetic series • Cross-Lesson: 1.01 Arithmetic & Number Sense

An arithmetic sequence has first term 3 and common difference 4. Find the sum of its first 15 terms.

$S = \frac{15}{2}(2 \times 3 + 14 \times 4) = 7.5 \times (6 + 56) = 7.5 \times 62 = 465$.

19 [Challenging] • Target: Sum of an arithmetic series • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Find the sum of the first 25 positive even integers ($2, 4, \ldots, 50$).

$a = 2$, $d = 2$, $n = 25$. $S = \frac{25}{2}(4 + 24 \times 2) = 12.5 \times 52 = 650$.

20 [Challenging] • Target: Special sequences • Cross-Lesson: 3.01 Algebra Essentials

The sequence $1, 4, 9, 16, 25, \ldots$ consists of perfect squares. Find the 8th term.

The nth term is n^2, so the 8th term is $8^2 = 64$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Geometric sequences

A geometric sequence has 3rd term 20 and 4th term 40. Find the first term and the common ratio.

a. $a_1 = 5, r = 2$ ✓

b. $a_1 = 20, r = 2$

c. $a_1 = 5, r = 4$

d. $a_1 = 10, r = 2$

$r = 40 \div 20 = 2$; $a_1 = 20 \div r^2 = 20 \div 4 = 5$.

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: MCQ — Special sequences

A sequence starts $3, 5, 8, 13, 21, \ldots$, where each term is the sum of the two terms before it. Find the 8th term.

a. 55

b. 89 ✓

c. 144

d. 76

Continuing: $3, 5, 8, 13, 21, 34, 55, 89$: the 8th term is 89.

23 [Foundational] • Target: MCQ — Visual and growing patterns

The bar chart shows the number of tiles used to build patterns 1 through 4. Predict the number of tiles in pattern 8.

a. 37

b. 40

c. 29

d. 33 ✓

$d = 4$, $a_1 = 5$: $T_8 = 5 + 7(4) = 33$.

24 [Foundational] • Target: MCQ — Visual and growing patterns

The bar chart shows the number of matchsticks used to build squares in a row, for 1 through 4 squares. Find the number needed for 12 squares.

a. 34

b. 43

c. 40

d. 37 ✓

$d = 3$, $a_1 = 4$: $T_{12} = 4 + 11(3) = 37$.

25 [Foundational] • Target: MCQ — Visual and growing patterns

The bar chart shows the number of matchsticks used to build hexagons in a row, for 1 through 4 hexagons. Find the number needed for 9 hexagons.

a. 41

b. 51

c. 50

d. 46 ✓

$d = 5$, $a_1 = 6$: $T_9 = 6 + 8(5) = 46$.

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: MCQ — Word problems

A theater has 18 seats in the first row, and each following row has 4 more seats than the row before. How many seats are in row 15?

a. 78

b. 70

c. 56

d. 74 ✓

$a_1 = 18$, $d = 4$: $T_{15} = 18 + 14(4) = 74$.

27 [Foundational] • Target: MCQ — Word problems

A ball is dropped and after each bounce rises to half its previous height. If it first rises to 120 cm after the first bounce, how high does it rise after the 5th bounce?

a. 60

b. 15

c. 3.75

d. 7.5 ✓

$a_1 = 120$, $r = 0.5$: $T_5 = 120(0.5)^4 = 7.5$ cm.

28 [Foundational] • Target: MCQ — Word problems

A savings plan deposits 80 riyals in week 1, then increases the deposit by 15 riyals each following week. Find the deposit in week 10.

a. 215 ✓

b. 200

c. 230

d. 935

$a_1 = 80$, $d = 15$: $T_{10} = 80 + 9(15) = 215$ riyals.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Word problems

A bacteria culture starts with 40 bacteria and triples every hour. How many bacteria are there after 4 hours?

a. 1,080

b. 3,240 ✓

c. 9,720

d. 480

$40 \times 3^4 = 40 \times 81 = 3{,}240$.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: MCQ — Pattern reasoning

The number of diagonals in a polygon with n sides is $\frac{n(n-3)}{2}$. Find the number of diagonals in a polygon with 9 sides.

a. 36

b. 27 ✓

c. 21

d. 54

$\frac{9(6)}{2} = 27$.

31 [Intermediate] • Target: MCQ — Pattern reasoning

A sequence follows the rule \"triple and subtract 2\", starting at 4 as the 1st term. Find the 4th term.

a. 34

b. 28

c. 82 ✓

d. 250

$4 \to 10 \to 28 \to 82$: the 4th term is 82.

32 [Intermediate] • Target: MCQ — Pattern reasoning

The nth term of a sequence is $T_n = 4n - 3$. Find the 15th term, and state whether the sequence is arithmetic or geometric.

a. 57, geometric

b. 53, arithmetic

c. 60, arithmetic

d. 57**, arithmetic** ✓

$T_{15} = 4(15) - 3 = 57$; since T_n is linear in n, it's arithmetic.

33 [Intermediate] • Target: MCQ — Identifying the type of sequence

Find the 3rd term of the sequence $-3, 6, -12, 24, \ldots$

a. 12

b. -10

c. -14

d. -12 ✓

$a = -3, r = -2$. $a_3 = -3 \times (-2)^2 = -3 \times 4 = -12$.

© ClipSAT — clipsat.org | All Rights Reserved

34 **[Intermediate]** • **Target: MCQ — Finding missing terms**

[illegible] An arithmetic sequence has first term 5 and common difference 6. What is the sum of its first 18 terms?

a. **1008** ✓

b. 990

c. 1098

d. 1026

$a_{18} = 5 + (18 - 1)(6) = 107, \quad S_{18} = \frac{18(5 + 107)}{2} = 1008$

35 **[Intermediate]** • **Target: MCQ — Sum of an arithmetic series**

[illegible] The arithmetic sequence follows the pattern: 3, 7, 11, 15, ... What is the fifth term in this sequence?

a. 17

b. **19** ✓

c. 23

d. 27

[illegible] 4 (7 - 3 = 4[illegible] 11 - 7 = 4[illegible]). [illegible] 3. [illegible] 4 [illegible] (15) [illegible] 4. [illegible]: 3 [illegible]: 3 + 4 = 7 [illegible]: 7 + 4 = 11 [illegible]: 11 + 4 = 15 [illegible]: 15 + 4 = 19 The given sequence is an arithmetic sequence where the common difference between consecutive terms is 4 (7 - 3 = 4, 11 - 7 = 4, etc.). The first term is 3. To find the fifth term, we.

© ClipSAT — clipsat.org | All Rights Reserved

36 **[Intermediate]** • **Target: MCQ — Sum of a geometric series**

■■■■ ■■■■■■■■■■ ■■■■■■■■■ a_n ■■■■■■ $5, 11, 17, 23, \ldots$. ■■■ ■■■ ■■■■■ $a_k = 119$■ ■■■■ ■■ ■■■■■ k■ The arithmetic sequence a_n follows the pattern $5, 11, 17, 23, \ldots$. If $a_k = 119$, what is the value of k?

a. 19

b. **20** ✓

c. 21

d. 22

■■■■■■■■■■ ■■■■■■■■ ■■ ■■■■■■■■ ■■■■■■■ ■■■■■ ■■■■■■ $a_1 = 5$ ■■■■■■■■ $d = 11 - 5 = 6$. The given sequence is an arithmetic sequence with first term $a_1 = 5$ and common difference $d = 11 - 5 = 6$. ■■■■ ■■■■■ ■■■■■■■ ■■■■■■■■■■ ■■■■■■■■■ ■■ $a_n = a_1 + (n - 1)d$. The formula for the nth term of an arithmetic sequence is $a_n = a_1 + (n - 1)d$. ■■■■ ■■■■■■ ■■■■■ k ■■■■■■ ■■■■■ $a_k = 119$. ■■■■■■■■■ ■■ ■■■■■■■: We want to find the value of k when $a_k = 119$. Substituting into the formula: $119 = 5 + (k - 1)6$ ■■■■ 5 ■■ ■■■■■■■■: S.

37 **[Intermediate]** • **Target: MCQ — Special sequences**

■■ ■■■■■■ ■■■■■■■■■■ ■■■■■■■■■ ■■■■■■■■■■■■: $18 + 6 + 2 + \frac{2}{3} + \ldots$ ■ What is the sum of the infinite geometric series: $18 + 6 + 2 + \frac{2}{3} + \ldots$?

a. **27 27** ✓

b. 26 26

c. 36 36

d. 24 24

■■■■■ ■■■■■■ 18■ ■■■■■■■■ 1/3. ■■■■■■■■ = 18 / (1 - 1/3) = 18 / (2/3) = 27. / First term 18, common ratio 1/3. Sum = 18 / (2/3) = 27.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: MCQ — Visual and growing patterns • Cross-Lesson: 1.01 Arithmetic & Number Sense

[illegible]: [illegible]. [illegible] x [illegible] 5 [illegible] x [illegible] 40 [illegible]

A grocery store plans to arrange display shelves for three types of fruit: apples, bananas, and oranges. If the number of apple boxes is x, the number of banana boxes is double the number of apple boxes, and the number of orange boxes is 5 less than the total number of apple and banana boxes combined, what is the number of orange boxes in terms of x, given that the total number of boxes for all fruits is 40?

a. $x - 5$

b. $3x - 5$ ✓

c. $2x - 5$

d. $4x - 5$

Let A be the number of apple boxes, B be the number of banana boxes, and O be the number of orange boxes. We are given that $A = x$. The number of banana boxes is double the number of apple boxes, so $B = 2x$. The number of orange boxes is 5 less than the total number of apple and banana boxes combined: $O = (A + B) - 5$. Substituting the expressions for A and B: $O = (x + 2x) - 5$. Simplifying this, we get $O = 3x - 5$. We are also given that the total number of boxes for all fruits is 40.

39 [Challenging] • Target: MCQ — Word problems • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

[illegible] 10 [illegible] 22. [illegible] a_1 [illegible] d [illegible] $a_1 + 3d$. In an arithmetic sequence, the third term is 10 and the seventh term is 22. If the first term is a_1 and the common difference is d, find the value of $a_1 + 3d$.

a. 14

b. **16** ✓

c. 18

d. 20

[illegible] $a_n = a_1 + (n-1)d$. The arithmetic sequence is given by the formula $a_n = a_1 + (n-1)d$. [illegible]: We have: $a_3 = a_1 + (3-1)d = a_1 + 2d = 10$ ([illegible] 1) $a_3 = a_1 + (3-1)d = a_1 + 2d = 10$ (Equation 1) $a_7 = a_1 + (7-1)d = a_1 + 6d = 22$ ([illegible] 2) $a_7 = a_1 + (7-1)d = a_1 + 6d = 22$ (Equation 2) [illegible] 1 [illegible] 2[illegible]: Subtracting Equation 1 from Equation 2, we get: $(a_1 + 6d) - (a_1 + 2d) = 22 - 10$ $4d = 12$ $d = 3$ $4d = 12$ $d = 3$ [illegible] d.

© ClipSAT — clipsat.org | All Rights Reserved

40 [Challenging] • Target: MCQ — Pattern reasoning • Cross-Lesson: 3.01 Algebra Essentials

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] 7 [illegible] [illegible] [illegible] [illegible] 19. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] x[illegible] [illegible] [illegible] x. We have an arithmetic sequence where the first term is 7 and the third term is 19. If the tenth term is x, find the value of x.

a. 49

b. 55

c. **52** ✓

d. 58

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] a_n [illegible] [illegible] [illegible] [illegible] a_1 [illegible] d [illegible]: $a_n = a_1 + (n-1)d$. [illegible] [illegible] $a_1 = 7$ [illegible] [illegible] $a_3 = 19$. [illegible] [illegible] [illegible] [illegible]: $a_3 = a_1 + (3-1)d$ $19 = 7 + 2d$ [illegible] 7 [illegible] [illegible]: $12 = 2d$ [illegible] [illegible] [illegible] 2: $d = 6$. [illegible] [illegible] [illegible] [illegible] 6. [illegible] [illegible] [illegible] [illegible] [illegible] a_{10}[illegible] [illegible] [illegible] [illegible] [illegible] x. [illegible] [illegible]: $a_{10} = a_1 + (10-1)d$ $x = 7 + (9 \times 6)$ $x = 7 + 54$ $x = 61$. In an arithmetic sequence, any term a_n can be written in terms of t.

41 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 4.01 Geometry & Measurement

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] 10 [illegible] [illegible] [illegible] [illegible] 22. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] a_1 [illegible] [illegible] d[illegible] [illegible] [illegible] [illegible] [illegible] a_1[illegible] In an arithmetic sequence, the third term is 10 and the seventh term is 22. If the first term is a_1 and the common difference is d, what is the value of a_1?

a. 2

b. 4

c. **6** ✓

d. 8

[illegible] [illegible] [illegible] [illegible] [illegible]: $a_n = a_1 + (n-1)d$. [illegible] [illegible]: $a_3 = 10$ => $a_1 + (3-1)d = 10$ => $a_1 + 2d = 10$ ([illegible] 1) $a_7 = 22$ => $a_1 + (7-1)d = 22$ => $a_1 + 6d = 22$ ([illegible] 2) [illegible] [illegible] 1 [illegible] [illegible] 2: $(a_1 + 6d) - (a_1 + 2d) = 22 - 10$ $4d = 12$ $d = 3$ [illegible] [illegible] d [illegible] [illegible] 1: $a_1 + 2(3) = 10$ $a_1 + 6 = 10$ $a_1 = 10 - 6$ $a_1 = 4$ [illegible] [illegible] [illegible] [illegible] a_1 [illegible] 4. We have the formula for an arithmetic sequence: $a_n = a_1 + (n-1)d$. The given information is: $a_3 = 10$ =>.

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Geometric sequences • Cross-Lesson: 5.01 Quantitative Comparisons

[illegible] u_n [illegible] $u_n = an + b$. [illegible] $u_3 = 10$ [illegible] $u_7 = 22$[illegible] [illegible] [illegible] b[illegible] [illegible] a [illegible] b [illegible] The arithmetic sequence u_n is defined by the formula $u_n = an + b$. If $u_3 = 10$ and $u_7 = 22$, what is the value of b, where a and b are constants?

a. 1

b. **2** ✓

c. 3

d. 4

[illegible] [illegible] $u_n = an + b$[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]: 1. $u_3 = 3a + b = 10$ 2. $u_7 = 7a + b = 22$ [illegible] [illegible] (1) [illegible] [illegible] (2)[illegible] [illegible] [illegible]: $(7a + b) - (3a + b) = 22 - 10$ $4a = 12$ $a = 3$ [illegible] [illegible] [illegible] a [illegible] [illegible] (1): $3(3) + b = 10$ $9 + b = 10$ $b = 10 - 9$ $b = 1$ [illegible] [illegible] b [illegible] 1. Since $u_n = an + b$, we get a system of two linear equations: 1. $u_3 = 3a + b = 10$ 2. $u_7 = 7a + b = 22$ Subtracting equation (1) from equation (2), we get: $(7a + b) - (3a + b) = 22 - 10$ $4a = 12$.

43 [Challenging] • Target: Identifying the type of sequence • Cross-Lesson: 7.01 Data Analysis & Probability

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [

44 [Challenging] • Target: Finding missing terms • Cross-Lesson: 8.01 Word Problems & Reasoning

■■ ■■ ■■■■■ ■■■■■■■ ■■ ■■■■■■■■■■ ■■■■■■■■■ ■■■■■■■■: 3■ 7■ 11■ 15■ ...■ What is the next term in the following arithmetic sequence: 3, 7, 11, 15, ...?

a. 17

b. 18

c. **19** ✓

d. 20

■■■■■■■■■■ ■■ ■■■■■■■■ ■■■■■■■ ■■■■ ■■■■■■ ■■■■ ■■ ■■■■■ ■■■■■■■■■ ■■■■■. ■■■■■■ = 7 - 3 = 4. ■■■■■■ = 11 - 7 = 4. ■■■■■■ = 15 - 11 = 4. ■■■■■ ■■■■■■ ■■■■■■■■ ■■ 4. ■■■■■■■ ■■■■■ ■■■■■■■■ ■■■■■ ■■■■■■ ■■■■■■■■ ■■■■ ■■■■■ ■■■■■■■. ■■■■■ ■■■■■■■ = 15 + 4 = 19. The sequence is an arithmetic sequence because the difference between consecutive terms is constant. Difference = 7 - 3 = 4. Difference = 11 - 7 = 4. Difference = 15 - 11 = 4. Thus, the common difference is 4. To find the next term, we add the common difference to the last term. Next te.

45 [Challenging] • Target: Sum of an arithmetic series • Cross-Lesson: 1.01 Arithmetic & Number Sense

■■■■■ ■■■■■■■■ ■■■■■■■ ■■■■■■ ■■■■■■ $a_1 = 5$ ■ ■■■■■■ ■■■■■■■■ $d = 3$. ■■ ■■ ■■■■■ ■■■■■ ■■■■■■■ a_5 ■■■■■ ■■■■■■■■■■■ An arithmetic sequence starts with the first term $a_1 = 5$ and a common difference $d = 3$. What is the value of the fifth term a_5 of this sequence?

a. **17** ✓

b. 20

c. 23

d. 26

■■■■■■ ■■■■■ ■■■■■ n ■■ ■■■■■■■■■■ ■■■■■■■■■ ■■ $a_n = a_1 + (n-1)d$. ■■ ■■■■ ■■■■■■■■ $a_1 = 5$ ■ $d = 3$. ■■■■■■■ ■■■■■ ■■■■■■■ a_5■ ■■■■■ $n = 5$: $a_5 = 5 + (5-1) \times 3$ $a_5 = 5 + (4) \times 3$ $a_5 = 5 + 12$ $a_5 = 17$ The formula for the n-th term of an arithmetic sequence is given by $a_n = a_1 + (n-1)d$. In this case, $a_1 = 5$ and $d = 3$. To find the fifth term a_5, we substitute $n = 5$: $a_5 = 5 + (5-1) \times 3$ $a_5 = 5 + (4) \times 3$ $a_5 = 5 + 12$ $a_5 = 17$

© ClipSAT — clipsat.org | All Rights Reserved